Gowtham Thangavelu

PLM SOLUTION ANALYST | DIGITAL ENGINEERING | DIGITAL THREAD & DIGITAL TWIN

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PROFESSIONAL SUMMARY

PLM and Digital Transformation Consultant with 8+ years of experience spanning mechanical engineering, PLM implementation, product data management, semiconductor product lifecycle, enterprise integration and digital engineering. Experienced across Siemens Teamcenter, 3DEXPERIENCE/ENOVIA, PTC Windchill, SAP PLM and Centric PLM, with hands-on experience in product structures, integrated BOMs, configuration and variant management, lifecycle and engineering change processes, and PLM process transformation. Combines engineering-domain depth with consulting and transformation experience to translate business capabilities into scalable PLM processes, govern product data and connect PLM with enterprise applications, enabling Digital Thread and Digital Twin outcomes.

ARCHITECTURE & CORE COMPETENCIES

PLM Architecture: PLM Solution Architecture | Business Capability Mapping | PLM Process Transformation | Future-State Roadmaps

Teamcenter: Teamcenter PLM | Product Structure | BOM Management | Lifecycle | Change Management | Configuration

Product Data: Product & Engineering MDM | Data Governance | Traceability | Integrated / Multi-Level BOM | Variant Management

Digital Engineering: Digital Thread | Digital Twin | Digital Engineering | R&D Transformation | New Product Introduction

Enterprise Integration: PLM–SAP Integration | SAP PLM | Enterprise Applications | Cloud-PLM Integration | Data Flows

Engineering Systems: 3DEXPERIENCE | ENOVIA | Windchill | NX | Creo | CATIA | SolidWorks | MCAD / PLM Alignment

Transformation: Business Workshops | Solution Consulting | Proof of Concepts | Process Standardization | Change Management

AI-Enabled Engineering: Writer AI | AI Agents | PLM Virtual Assistant | Enterprise Connectors | Skill .md Development | Workflow Automation

Industry: Semiconductor | Automotive | Consumer Products | Heavy Machinery

PROFESSIONAL EXPERIENCE

PLM Digital Operations Analyst | e.l.f. Beauty | ***Dec 2025 – Present***

• Architected PLM process and product-data models within Centric PLM, translating product lifecycle requirements into scalable business processes spanning product definition, configuration, BOM management, Trigger Readiness, SKU readiness, supplier processes, and downstream enterprise handoffs.

• Engineered complex product master-data structures across Products, Colorways, SKUs, Supplier Quotes, BOMs, and product attributes, including consolidation of seasonal launch views into unified annual brand architectures to improve data consistency, lifecycle visibility, and downstream consumption.

• Defined PLM data governance and information-management patterns covering data ownership, attribute structures, validation controls, lifecycle consistency, traceability, and cross-system data integrity across product development processes.

• Designed cross-system integration architectures connecting Centric PLM with SAP S/4HANA, Snowflake, and regulatory platforms such as Ithos, defining API-based data flows, JSON payload/data mapping, integration boundaries, and middleware orchestration using MuleSoft and SAP BTP Integration Suite / CPI.

• Orchestrated end-to-end product-data flows across PLM and enterprise applications, aligning business processes, system-of-record responsibilities, integration dependencies, and downstream data consumption to establish a scalable Digital Thread foundation.

• Engineered integration and solution validation frameworks covering end-to-end business scenarios, data mapping and reconciliation, REST/API validation, performance testing, defect analysis, retesting, and Pre-Production/Production readiness.

• Steered cross-functional product-data governance across Product Development, Sourcing, Supply Chain, IT, Integration, SAP, Data, and regulatory stakeholders, establishing consistent process ownership, configuration integrity, data quality, and downstream usability.

• Defined future-state PLM and Digital Thread architecture opportunities spanning process standardization, automation, data-model optimization, enterprise-system alignment, and scalable integration patterns supporting future Digital Twin capabilities.

• Applied Writer AI daily as an enterprise productivity and automation platform, leveraging connectors and AI agents to streamline PLM knowledge access, process execution, and business-user enablement.

• Designed reusable Skill .md assets for PLM and regulatory platforms, converting complex system knowledge and process guidance into AI-accessible capabilities that help business users resolve queries, understand processes, and reduce manual effort.

• Designed a PLM Virtual Assistant using Writer AI agents and enterprise connectors, enabling conversational access to PLM-related knowledge and guidance for business users.

MES MOM Specialist | PLM Business Transformation Consultant | Accenture | Nov 2024 – Dec 2025

• Architected PLM transformation solutions and business capabilities across Engineering and R&D organizations, translating product lifecycle requirements into solution patterns spanning Product Definition, Integrated BOM, Product & Engineering Master Data Management, Digital Thread, and Digital Twin.

• Designed PLM functional solution frameworks covering product/package design, part management, BOM management, Engineering Change Management, document management, product structures, research, simulation, prototyping, product testing, and supplier collaboration.

• Engineered Integrated BOM and product-data management approaches to establish consistent product structures, improve information traceability, and connect engineering processes across the product lifecycle.

• Defined future-state PLM transformation roadmaps, assets, proof-of-concepts, methodologies, and solution offerings, aligning PLM capabilities with business transformation objectives and enterprise operating models.

• Architected Digital Thread and Digital Twin use cases by connecting product data, engineering processes, manufacturing readiness, lifecycle information, and emerging digital engineering capabilities.

• Steered PLM solution development and senior stakeholder engagement, translating complex product-data and process requirements into transformation propositions, solution concepts, PoVs, and go-to-market offerings.

• Engineered PLM modernization approaches spanning cloud integration, product-data management, integrated BOMs, process standardization, and scalable digital engineering capabilities, with emphasis on maintainability and long-term enterprise value.

Senior Engineer | Bosch Global Software Technologies | Sep 2022 – Nov 2024

- Supported Siemens Teamcenter PLM implementation and global PLM transformation initiatives, translating engineering and business needs into PLM process and capability improvements.
- Represented Bosch India as Country Approval Product Manager (CAPM) in worldwide Teamcenter PLM transformation forums, contributing to cross-country alignment and solution standardization.
- Hands-on experience with Teamcenter change management and component lifecycle activities, supporting governance of engineering product information and lifecycle processes.
- Created and enhanced PLM enablement and BOM-management assets; assessed current-state processes and identified improvement opportunities through PLM adoption and transformation.
- Delivered technology guidance, proof-of-concepts, business workshops and user training while applying PLM industry best practices.
- Contributed to Digital Thread/Digital Twin, cloud-PLM, graph-based product data and SaaS PLM initiatives, with focus on scalable and maintainable PLM approaches.
- Applied Design for Excellence, DFMEA, 5-Why and Fishbone analysis and supported design-to-manufacturing transition, compliance and product lifecycle activities.

Consultant | Capgemini | Client: Lam Research | ***Sep 2021 – Sep 2022***

- Supported semiconductor digital manufacturing and Digital Twin initiatives, creating virtual representations of parts and layouts to improve operational efficiency, training and serviceability.
- Performed product configuration and variant-management activities and supported BOM and change-management processes across the 3DEXPERIENCE platform.
- Managed complex multi-level semiconductor BOMs of approximately 100,000 components and large assemblies of approximately 50,000 components, supporting product structure, configuration and lifecycle analysis.
- Performed UAT for SAP PLM and 3DEXPERIENCE and analyzed SAP, Teamcenter and CAD/model data to support product and aftermarket use cases.
- Created and released engineering changes and BOMs through ECO processes; maintained top-level assemblies and supported product-data traceability across connected systems.
- Analyzed SAP BOM data to identify serviceable/non-serviceable parts and critical spares, supporting maintenance efficiency and aftermarket decision-making.

Design Engineer | Milacron India Private Limited | ***Jun 2019 – Sep 2021***

- Coordinated engineering projects for Milacron USA covering change management, BOM management and mechanical design development.
- Translated project inputs into work-flow sequences, coordinated engineering resources and quality checks, and managed delivery against project timelines.
- Created and revised solid/sheet-metal models, detailed drawings and BOMs; incorporated engineering changes and resolved assembly fit/interference issues with customers.

Design Engineer | Hirotec India Private Limited | Client: General Motors Tech Centre India | ***Feb 2018 – Jun 2019***

- Designed and detailed BIW welding fixtures using Siemens NX, including layouts, key sheets, tooling BOMs and revisions based on customer engineering inputs.
- Coordinated customer inputs, engineering resources and project schedules while working to NAAMS and GMNA standards.

CERTIFICATIONS

• Teamcenter PLM Development & Customization – Siemens Certified

• Project Management Foundations – PMI Certified

• Writer AI – Passport Pro | AI-Enabled Enterprise Transformation

EDUCATION

B.E. Mechanical Engineering | SNS College of Technology, Coimbatore | 2013–2017 | CGPA: 8.60

TECHNOLOGY LANDSCAPE

Teamcenter | 3DEXPERIENCE | ENOVIA | Windchill | SAP PLM | Centric PLM | NX | Creo | CATIA | SolidWorks | SAP | CPI / Integration | Jira | Confluence | Asana